entergy

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RBG-48175

10 CFR 50.73

June 9, 2022

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2022-02-00, "Condition Prohibited by Technical Specifications due to Reactor Scram and Turbine Trip Level 8 Trip Setpoint Deviation"

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments.

Should you have any questions, please contact Mr. Tim Schenk, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,

Bonnie Bryant

BLB/dmw

Enclosure: Licensee Event Report 50-458 / 2022-02-00, "Condition Prohibited by Technical Specifications due to Reactor Scram and Turbine Trip Level 8 Trip Setpoint Deviation"

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station
NRC Project Manager - River Bend Station

Enclosure

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NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023
LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

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4. Title
Condition Prohibited by Technical Specifications due to Reactor Scram and Turbine Trip Level 8 Trip Setpoint Deviation

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name N/A	Docket Number 05000 N/A
04	13	2022	2022	-002 -	00	06	09	2022	Facility Name N/A	Docket Number 05000 N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include Area Code)
Tim Schenk, Manager Regulatory Assurance	225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ **Yes** (If yes, complete 15. Expected Submission Date)		N/A	N/A	N/A

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On July 15, 2021, with River Bend Station (RBS) operating at 100% power, an RBS Safety Analysis review identified that when accounting for transmitter drift and other uncertainties, the original station setpoints for reactor water level 8 reactor scram and the original station setpoints for reactor water level 8 turbine trip could allow a turbine trip to occur prior to the reactor scram resulting in an over pressurization event.

RBS implemented an interim penalty to Operating Limit Minimum Critical Power Ratio (OLMCPR) on July 22, 2021. A past operability evaluation completed on April 13, 2022, concluded that the OLMCPR Technical Specification was exceeded on multiple occasions during the past two operating cycles. The cause of this condition was a result of not considering instrument drift and uncertainty during the original development of the Level 8 reactor scram and turbine trip setpoints for RBS.

Corrective actions include implementing an administrative limit to OLMCPR that is applied to all rated and off-rated conditions across all Core Operating Limit Report (COLR) Application Conditions, issuing an Operating Experience (OE) Bulletin to reinforce the behaviors necessary when receiving fleet and industry OE, and raising the turbine trip nominal trip setpoint.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
River Bend Station, Unit 1	05000 - 458	2022	- 002	- 00

NARRATIVE

EVENT DESCRIPTION

On July 15, 2021, with River Bend Station (RBS) operating at 100% power, a potential condition was identified with the Reactor Pressure Vessel [RPV] level 8 instrumentation. There are different sets of level 8 instrumentation loops associated with the reactor water level 8 reactor scram and the reactor water level 8 turbine trip. The nominal trip setpoint (NTSP) for reactor water level 8 reactor scram is 51.0 inches and the NTSP reactor water level 8 turbine trip is 50.7 inches. Since there are separate sets of transmitters for these trips, it is theoretically possible for each set of level transmitters (LT) to see the effects of drift and other uncertainties. Failure of the level 8 reactor scram initiation prior to or simultaneous with a turbine trip could lead to a severe Minimum Critical Power Ratio (MCPR) transient and the expected MCPR would not support the current limits. RBS accident analysis input and vendor model assumes simultaneous actuation.

The limiting transient event, specifically the Feedwater Controller Failure (FWCF) transient, was re-evaluated at rated power conditions for RBS Cycle 22 with input from Global Nuclear Fuels (GNF). This evaluation used the assumption that the turbine trip would occur prior to the level 8 reactor scram. The evaluation resulted in GNF recommending a 0.05 OLMCPR penalty. On August 19, 2021, RBS issued Standing Order 363 Revision 4 to include the OLMCPR penalty resulting from the level 8 issue combined with the OLMCPR penalty that previously resulted from Safety Communication 21-04 (SC 21-04) (reference RBS LER 21-004).

A past operability evaluation was performed to determine if any MCPR values were exceeded in the past three years from July 15, 2018 – July 15, 2021. The past operability evaluation was performed with the same assumption that the turbine trip would occur prior to the level 8 reactor scram, while concurrently applying penalties resulting from both SC 21-04 and the level 8 issue. The past operability completed on April 13, 2022, concluded that OLMCPR limits were exceeded for a time longer than allowed by Technical Specification 3.2.2 on multiple dates in Cycle 20 and Cycle 21 due to the applied penalties. During Cycle 22, OLMCPR limits were not exceeded for a time longer than allowed by Technical Specification 3.2.2.

This condition is being reported as an operation prohibited by Technical Specifications in accordance with 10 CFR 50.73(a)(2)(i)(B).

SAFETY ASSESSMENT

The limiting transient event, specifically the response of the FWCF transient, has not occurred while the RBS OLMCPR was exceeded due to the applied penalties resulting from SC 21-04 and the level 8 issue. It can be concluded that the Safety Limit Minimum Critical Power Ratio (SLMCPR) was not violated.

Further, regarding the physical condition of the fuel observed during subsequent refuel outages, no fuel damage occurred due to this condition. Fuel operating in the RBS core across all three operating cycles has undergone either in-mast or vacuum canister sipping, and all fuel failures discovered during this time were determined to be due to debris fretting.

The condition did not prevent a safety function, as no limiting transient event as analyzed in the RBS reload evaluation occurred while the OLMCPR was exceeded due to the penalties applied. As a result, there was no impact to the health and safety of the public or plant personnel from this condition.

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EVENT CAUSE

When the level 8 trip setpoints were originally determined for RBS, the setpoints were not evaluated to consider drift and uncertainty to ensure the reactor scram would occur before or simultaneously with the turbine trip to maintain alignment with the current safety analysis.

RBS did not adequately incorporate fleet operating experience from Grand Gulf Nuclear Station (GGNS) when it was originally identified in CR-GGNS-2000-1810.

CORRECTIVE ACTIONS

Complete:
- RBS applied the vendor recommended administrative limits to OLMCPR on August 19, 2021.
- RBS work order 569699 to adjust the level 8 turbine trip setpoint has been created.

Planned Actions tracked in the Corrective Action Program:
- RBS work order 569699 to adjust the level 8 turbine trip setpoint has been added to the outage worklist and is being tracked in the corrective action process.
- RBS will issue an OE Bulletin to reinforce the behaviors necessary when receiving fleet and industry OE. This item is also being tracked in the corrective action process.

PREVIOUS SIMILAR EVENTS

None